

EXPERIMENT 13:

Construct a C program for implementation of the various memory allocation Strategies

```
1 #include <stdio.h>
2
3 void bestfit(int mp[], int p[], int m, int n) {
4     int j = 0;
5     for (int i = 0; i < n; i++) {
6         if (j >= m) {
7             break;
8         }
9         if (mp[i] >= p[j]) {
10             printf("\n%d fits in %d", p[j], mp[i]);
11             mp[i] = mp[i] - p[j];
12             j++;
13             i = i - 1;
14         }
15     }
16     for (int i = j; i < m; i++) {
17         printf("\n%d must wait for its process", p[i]);
18     }
19     printf("\n");
20 }
21
22 void rsort(int a[], int n) {
23     for (int i = 0; i < n; i++) {
24         for (int j = 0; j < n; j++) {
25             if (a[i] > a[j]) {
26                 int t = a[i];
27                 a[i] = a[j];
28                 a[j] = t;
29             }
30         }
31     }
32 }
33
34 void sort(int a[], int n) {
35     for (int i = 0; i < n; i++) {
36         for (int j = 0; j < n; j++) {
37             if (a[i] < a[j]) {
38                 int t = a[i];
39                 a[i] = a[j];
40                 a[j] = t;
41             }
42         }
43     }
44 }
45
46 void firstfit(int mp[], int p[], int m, int n) {
47     sort(mp, n);
48     sort(p, m);
49     bestfit(mp, p, m, n);
50 }
```

Status: **Accepted**

Memory Used: 728 byte Execution Time: 0.003 sec

Number of memory partition : Number of process : Enter the memory partitions :
Enter process size :
1. Firstfit 2. Bestfit 3. worstfit
Enter your choice :
212 fits in 500
417 must wait for its process
112 must wait for its process

Input

3
3
100
500
200
212
417
112
1